Lab 1

Introduction to Databases

What is Database?

- Three levels to view:

Level 1: literal meaning – the place where data is stored

Database = Data + Base, the actual storage of all the information that are interested

Database properties:

- It represents some aspect of the real world
- It is a logically coherent collection of data with inherent meaning
- It is designed, built and populated with data for specific purpose
- A Database can be of any size ranging from few number of records have hundreds or thousands bytes to gigabytes or terabytes
- A Database may be generated and managed manually or it may be computerized
- The computerized choice may be by a group of application programs implemented specifically for that application or by a:

Database Management System (DBMS)

Advantages of Database

A database system offers several key benefits that address the limitations of traditional file based systems and enhance overall data management efficiency:

Reduces Data Redundancy
Minimizes duplication of data by storing it centrally, ensuring that each piece of information exists only once across the system.

Controls Data Inconsistency
Maintains uniform and accurate data by ensuring that any update is reflected consistently throughout all related records.

Facilitates Data Sharing
Allows multiple users and applications to access and use the same data simultaneously, supporting collaboration and integration.

Enforces Data Standards
Promotes uniformity by applying consistent naming conventions, formats, and rules across the entire database.

Enhances Data Security
Protects sensitive information through user authentication, access controls, and permission management, allowing only authorized access.

Maintains Data Integrity
Ensures accuracy and reliability of data using constraints (like primary and foreign keys) and referential integrity rules.

Improves System Performance and Efficiency
Provides faster data access, better resource utilization, and smoother data operations through centralized management and optimization.

Disadvantages of Database

While database systems provide numerous benefits, they also introduce certain challenges and limitations that must be managed carefully:

Security Risks
Without strong access controls and monitoring, sensitive data can be exposed to unauthorized users, leading to potential data breaches.

Integrity Challenges
If constraints and validation rules are not properly enforced, data accuracy and consistency may be compromised.

High Hardware Requirements
Large and complex databases often require additional storage capacity and powerful servers to operate efficiently.

Performance Overhead
Complex queries, large data volumes, or multiple concurrent users can slow down system performance and increase processing time.

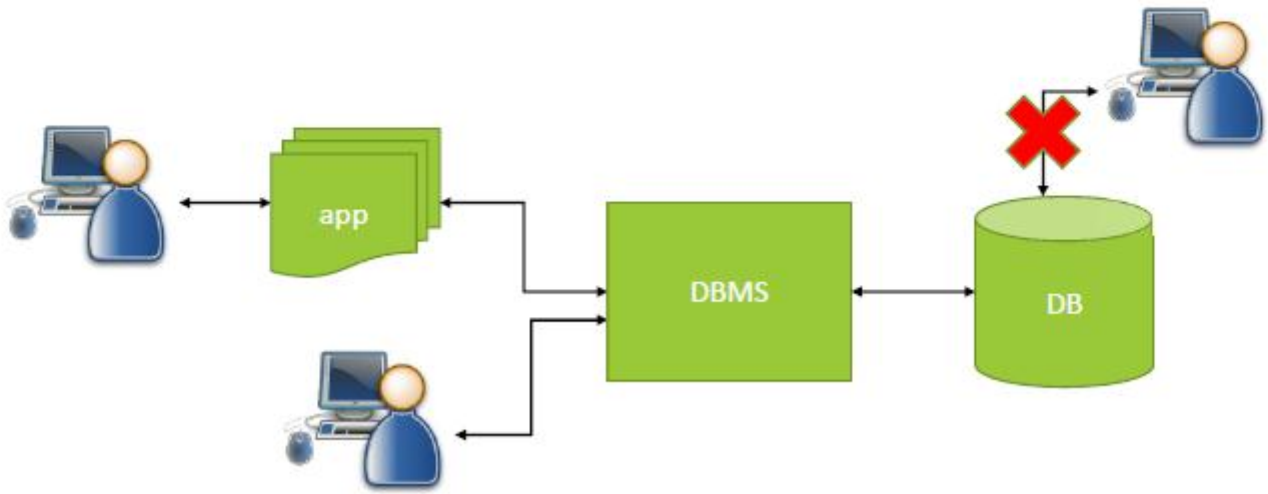
System Complexity
Database management systems are inherently complex, requiring skilled administrators and developers to configure, optimize, and maintain them effectively.

Level 2: Database Management System (DBMS)

The software tool package that helps gatekeeper and manage data storage, access and maintenances. It can be either in personal usage scope (MS Access, SQLite) or enterprise level scope (**Oracle**, MySQL, MS SQL, etc).

Level 3: Database Application

All the possible applications built upon the data stored in databases (web site, BI application, ERP etc).



What is Oracle Database 10g?

- Oracle Database 10g is a **Relational Database Management System (RDBMS)** developed by Oracle.

- It is designed to **store, manage, retrieve, and secure large amounts of data** efficiently.

- It supports both:

Relational model (RDBMS) → tables, rows, columns
Object-Relational model (ORDBMS) → objects, complex data types, multimedia

Main Components of Oracle 10g (Grid Infrastructure)

- Oracle 10g is built around **Grid Computing**, which means computing resources work like a shared utility.

- The three main products are:

Oracle Database 10g

Stores enterprise data
Supports structured & unstructured data
Uses SQL and PL/SQL
Provides security, reliability, and scalability

Oracle Application Server 10g

Runs business applications
Supports web apps, portals, business intelligence
Handles business logic

Oracle Enterprise Manager 10g Grid Control

Manages and monitors databases and servers
Automates installation and configuration
Monitors performance and availability

Features of Oracle Database

Oracle Database is loaded with features and is **powerful**, **feature-rich**, **scalable**, and **secure** enough for the business requirements of today. It thus best fits the application to handle **massive** applications with real-time data effectively. Some of the salient features include high availability features, superior performance, in-depth analytics, with robust data security.

Components of Oracle Database

The components of Oracle Database include **Oracle Instance, Database, Tables paces, Schemas, Processes, Redo Logs, Control Files,** and **Oracle Net**. Each one of them, in coordination with another carries out the work of management, storage, and effective retrieval of data. hence improves the performance of the overall database.

Instance: Manages memory (SGA, PGA) and processes that interact with the database.

Database: Contains physical files like data files (store data), control files (track the database), and redo logs (for recovery).

Tables paces: Logical units that organize data storage in the database.

Schemas: Collections of database objects (tables, views) owned by a specific user.

Data Dictionary: Stores metadata, including user information, privileges, and object definitions.

Processes: Includes background processes like DBWn (writes data), LGWR (logs changes), SMON (system recovery), and PMON (process cleanup).

Redo Logs: Log changes for use in database recovery after failures.

Control Files: Maintain the structure of the database and track log information.

Oracle Net: Enables communication between the database server and client over a network.

Editions of Oracle database

Oracle Database offers different editions to suit various needs.

Oracle Standard Edition (SE): It is Suitable for small to medium businesses because of its core database functionality and more affordablity.

Oracle Enterprise Edition (EE): This edition is for large organizations that need advanced features like RAC (Real Application Clusters), data partitioning, and high security. It's ideal for important, mission-critical applications.

Oracle Express Edition (XE): It is free but very limited by CPU and storage. Mainly for students, developers and small projects.

Oracle Cloud Edition: This database targets cloud environments with maximum scalability and flexibility in Oracle infrastructures.

Oracle Lite: It is a very lightweight, targeted at mobile and embedded applications. It is designed for working at top gear both on resource-constrained mobile devices and at remote location implementations.

Oracle Express Edition (XE)

Oracle Database XE is easy to install. Oracle Database XE provides an Oracle database and tools for managing the database.

For download the program:

https://www.oracle.com/database/technologies/database-11g-express-edition.html